

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int pages[30], frame[10];
```

```
    int n, f, i, j, k = 0, fault = 0, hit;
```

```
    printf("Enter number of pages: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter reference string:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &pages[i]);
```

```
    printf("Enter number of frames: ");
```

```
    scanf("%d", &f);
```

```
    for(i = 0; i < f; i++)
```

```
        frame[i] = -1;
```

```
    printf("\nPage\t");
```

```
    for(i = 0; i < f; i++)
```

```
        printf("F%d\t", i + 1);
```

```
    printf("Status\n");
```

```
for(i = 0; i < n; i++)
{
    hit = 0;

    for(j = 0; j < f; j++)
    {
        if(frame[j] == pages[i])
        {
            hit = 1;
            break;
        }
    }

    if(hit == 0)
    {
        frame[k] = pages[i];
        k = (k + 1) % f;
        fault++;
    }

    printf("%d\t", pages[i]);

    for(j = 0; j < f; j++)
    {
        if(frame[j] == -1)
```

```
        printf("-\t");
    else
        printf("%d\t", frame[j]);
}

if(hit)
    printf("Hit\n");
else
    printf("Fault\n");
}

printf("\nTotal Page Faults = %d\n", fault);

return 0;
}
```